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SPectroscopic Absorption, Real Time Analysis

Concatenator · Absorbance Calculator · Plotter

Quick start guide

Getting data in, teaching the tool your file names,
and the recipes for the outputs you actually need.

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NSLS-II 22-IR-1 — Dr. Lee's Lab

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The same text is in the program: left panel, Guide / notes, View dropdown. Hover any control for a tip; F1 lists the keyboard shortcuts.

20 pages

QUICK START

This is the ten-minute path from raw segment files to a finished figure. Every step is reversible. The tool writes to disk when you press Run or Save.

For a first look, point the Input folder at demo_data, which ships beside the program. It holds a small synthetic series: 5 series points, 3 channels, 2 grating segments. It uses the classic 22-IR-1 naming convention, so every step below works on it unchanged, fringes and all. One of the five points carries a _D tag. Both the compression branch and the decompression branch are present. Every file in demo_data is synthetic.

About > 'Welcome & tour...' loads that folder and walks this path in the window itself. The program dims around the control it explains. Esc leaves the tour at any point.

1. POINT AT THE DATA

Left panel, Input folder: press Browse, or drag a folder from Explorer onto the window, or paste a path and press Enter. Enter also rescans and remembers the folder.

The small arrow beside Browse drops down your last 5 input folders plus 'Open in Explorer'. A right-click inside the box does the same.

The scan reads one folder. Point at the folder that holds the segment files themselves. If you pick a parent folder, the log says so and lists the subfolders it found.

Output folder: press Browse and pick anywhere you can write. A Run creates <output>/<inputname>_absorbance/ and puts the CSVs there. If that subfolder already exists and holds files, the tool appends a timestamp (<inputname>_absorbance_YYYYMMDD_HHMM). An earlier run keeps its own folder.

2. CHECK HOW THE NAMES ARE READ

The button under the Input box shows the active profile. The default profile is '22-IR-1 default'. That profile reads

```
vis_{DAC}_{Sample}[_{Pressure}][_bg|_s][_C|_D][_rep][.{seq}]
```

For example, vis_Y04_Arch29_26p0_s.003 is DAC Y04, sample Arch29, 26.0 GPa, the sample channel, grating segment 3.

If your filenames look like that, go to step 3.

For any other scheme, open the button. Press 'Guess format'. Check the Preview: green means parsed, red means skipped with the reason. Correct anything wrong, then press 'Use this profile'. The Naming system view holds the full walkthrough.

3. RUN

Press Run. The tool concatenates the grating segments of each measurement and computes

$$A = -\log_{10}[(\text{Sample} - \text{Dark}) / (\text{Background} - \text{Dark})]$$

The tool writes one CSV per measurement into the output subfolder. It also writes a `_reduction.provenance.json` sidecar, which records the tool version, the parameters and every curve.

Watch the Progress log. Each line is one measurement. OK lines carry the point count and the file written. SKIP lines carry the filename and the reason. Run turns into Cancel while a run is in flight.

Absorbance needs all three channels. A measurement with only some of them loads as raw counts. The tool tags the trace name ([S only], [S+B]).

4. LOOK AT THE DATA BEFORE YOU TRUST IT

Every trace plots at once. Do two checks first.

Inspect one trace (Plot tab > Plot mode). Pick a measurement. The left axis shows Sample, Background and Dark counts. The right axis shows its Absorbance. In a healthy point the background sits comfortably above the sample, and both channels sit under the detector ceiling.

Traces (Data tab). One row per loaded point. A colored dot beside a row means a quick quality check fired. Hover the dot for the reason (saturation, negative absorbance, missing channels).

Click a curve on the plot to select it. Double-click to solo it.

Click a legend entry to hide it. Right-click a curve for the quick-actions menu (inspect, solo, hide, toggle D, defringe compare, show in data table).

Ctrl+D opens the data table under the plot. It holds the numbers for the selected trace, with 'Copy all (TSV)' and 'Open in

Excel'.

5. MAKE IT A FIGURE

Export tab > Figure: pick a Journal preset. One pick sets the column width and the house style: typeface, text sizes, line weight, thin spines, ticks in, DPI. Tick 'Preview at export size (WYSIWYG)' in Export tab > Export to see the true printed proportions on screen.

Style tab > Legend: above about ten traces, turn on 'Direct labels at curves' or use Style tab > Colorbar. A legend box that size covers the data.

Export tab > Export: 'Save plot...' writes PNG / PDF / SVG / EPS / TIFF. PDF and SVG are vector. Leave 'Editable text' on to keep the labels editable in Illustrator or Inkscape.

WHAT ELSE IS WORTH KNOWING ON DAY ONE

Session tabs sit above the plot. Each tab is an independent session with its own data, folders, settings and undo history. Ctrl+T opens a blank one, Ctrl+W closes, Ctrl+Tab cycles, double-click renames. Use them to compare two loads side by side.

'Load previous run' reopens a finished output folder at once. The tool re-imports the CSVs for plotting, straight off disk. A blank tab also lists recent runs.

Rescan (or F5) re-runs the folder when new files appeared since the last Run. It is the between-measurements top-up. The Auto rescan pill in Plot tab > Plot mode does it unattended every N seconds.

Hover any control for a tip. Helper tips, in the top bar's Settings panel, is the master switch. F1 lists the keyboard shortcuts.

Theme sits on the top bar. The six themes above the divider are the working ones. They include High Contrast, Colorblind Safe and Colorblind Safe Dark. The rest of the list changes colors only.

The gear beside Theme opens the Settings panel. It holds Font, Text size, Helper tips, Performance mode, Tutorial and About. Font sets the typeface of every control. Pick OpenDyslexic

there for the dyslexia-friendly face.

The Quick Access strip above the tabs is yours to furnish. The gear in its corner opens a two-column checklist. CONTROLS on the left are settings the strip can carry a second copy of.

ESSENTIAL FUNCTIONS on the right are one-press actions: Run, Rescan, Open output, Data table, Save plot, Copy figure and more. They arrive as small labelled buttons under a rule. The strip starts empty. With 'Collapse all' pressed, section headers become handles: drag one up or down to rearrange its tab.

The tool remembers folders, theme, window size, default colormap, notes and presets between launches.

NUKE (top bar, center) is the hard reset. It clears the session: data, plot, folders, log, every control. It leaves your files on disk untouched. Saved presets and your default colormap survive. It asks first.

THE NAMING SYSTEM

How a filename becomes a measurement. Everything downstream is decided here: grouping, absorbance, legends, and output file names. Ten minutes with this view covers every folder you will meet.

To reduce a folder, Run has to know five things about every file. Which DAC it came from. Which sample. What the value of the experiment variable was. Which channel it is (sample / background / dark). Which grating segment it is. Those five are the whole job.

TOKENS

The tool reads a filename as a list of pieces. A piece, or token, is whatever sits between two separators.

```
vis_Y04_Arch29_26p0_s.003
 \ / \ / \ / \ / \ / \ /
 1   2   3   4   5 segment suffix
```

Token 1 is a fixed prefix. Tokens 2 and 3 are the DAC id and the sample id. Token 4 is the value and token 5 is the channel keyword. The '.003' at the end is the segment suffix. The tool splits it off the end of the name BEFORE it tokenizes anything.

The tool drops empty tokens, so a doubled separator reads as one.

A profile assigns a MEANING to each token position, in order. The meanings available are:

dac	the cell id. Required, unless a default supplies it.
sample	the sample id. Required, unless a default supplies it.
pressure	the value of the Series variable. Optional.
role	the channel keyword (sample / background / dark).
branch	the compression / decompression tag.
rep	the retake number.
ignore	a piece that carries nothing the pipeline needs.

'pressure' is the internal name of the field, whatever the Series variable is called. The dialog shows the variable's own name on the label and the preview column.

THE TWO KINDS OF PROFILE

22-IR-1 default (built-in). A fixed, hand-written grammar. It ships read-only, and its behavior holds across releases.

```
vis_{DAC}_{Sample}[_{Pressure}][_bg|_s][_C|_D][_rep][.{seq}]
```

- the name starts with 'vis' and carries at least three underscore-separated tokens
- a bare name = dark, _bg = background, _s = sample
- _C / _D is the branch tag; _2 / _3 is a retake. They may appear in either order
- the value uses 'p' for the decimal: 1p39 = 1.39. 0 is allowed. The tool reads a missing value token as 0 and notes it in the log
- the segment suffix is a dot plus digits (.001 .. .00N). Any number of segments is fine. Omit it for a single-stitch file and the segment is 1
- a trailing '.csv' twin is tolerated and deduplicated

Worked example:

```
vis_Y04_Arch29_26p0_s.003
DAC      Y04
sample   Arch29
value    26.0
channel  sample
branch   none
retake   1
segment  3
```

Rejections this grammar produces:

```
vis_Y04_Arch29_26p0_s_r.003
'unrecognized trailing token'. _r is not a known suffix. The
built-in accepts bg / s / C / D / a digit retake.
vis_Y04_Arch29_26p0_sA.003
The same reason. 'sA' is not the sample keyword.
vis_Y04_Arch29_bg.001.002
'malformed (extra extension)'. The base still holds a dot
after the tool split the segment off.
vis_Y04_Arch29_26p0_s.abc
'segment is not numeric'.
Y04_Arch29_26p0_s.003
'does not match vis_DAC_SAMPLE[_PRESSURE]'. There is no 'vis'
prefix.
```

Each of these has a fix that leaves your file names alone.
Teach a custom profile, or fix the individual files by hand.
Both are below.

Custom profiles. Everything above becomes editable. The tool saves a custom profile under a name, lists it in the Profile dropdown, and keeps it across restarts.

THE PROFILE FIELDS

Prefix

A fixed first token at the start of every filename. The tool compares it case-insensitively and then consumes it. The classic names use 'vis'. Blank suits names that open on a field. A name that starts with something else is skipped with "missing prefix '<x>'".

Separator

The text between tokens. It is usually `_` or `-`. Comma-separate alternatives to accept several: `'_,-'` splits on either. The tool tries alternatives longest-first, so a multi-character alternative wins over a single character inside it. A bare `'` means the comma itself is the separator.

Value decimal (labelled with the Series variable's name)

The character that stands for the decimal point inside the value token. `'p'` reads 12p5 as 12.5. `'.'` reads 12.5 directly. `','` reads 12,5. The tool also accepts `'.'` and `','` whatever you pick, so a folder that mixes 12p5 and 12.5 still parses.

Strip units

Unit text the tool removes from the END of the value token before it reads the number, comma-separated. With `'gpa,kbar'`, the token `'15.3GPa'` reads as 15.3. Matching is case-insensitive and the first match wins.

Segment sep

The text immediately before the grating-segment number at the very end of the name. It may be several characters: `'_seg'` reads `name_seg003`. The RIGHTMOST occurrence splits base from segment. Blank suits a convention of plain names, where every file is a single stitch.

Numbering

digits - any all-digit run, read with `int()`, so any padding works: `.001`, `.01` and `.1` all mean segment 1.
letters - `a = 1`, `b = 2 ... z = 26`, then `aa = 27` (spreadsheet style), case-insensitive. Capped at two letters, so a plain data extension (`.txt`, `.dat`, `.asc`) stays an extension.

No number =

What a bare name means. Give a segment index, normally 1, which means one stitch called segment 1. Give the word 'reject' to skip such files with the reason "no segment suffix ('<sep><segment>' required)". Use 'reject' when every real file in the convention is numbered, so a bare

name is something else: a log, a note, a stray export. 'reject' needs a segment separator, and the validator says so when the separator box is blank.

Background / Sample / Dark keyword(s)

The token that marks each channel. Comma-separate alternatives (bg,ref). Matching is case-insensitive. Dark is the default role, so the Dark box can stay empty. Fill it when your names say 'dark' explicitly.

Compression / Decompression keyword(s)

The branch tag, shown as C and D on the plot. D draws dashed. Classic names use c and d. Comma-separate alternatives. Leave both blank and the tool uses the classic c / d.

Default DAC Name / Default Sample Name

For single-cell folders whose names omit that piece. Put the value here AND drop that label from the token order. Set the chip to 'ignore', or leave the field chipless. dac and sample are the two required fields, and a token or a default supplies each.

Token order

The chips under 'Teach by example' set it. It is the list of meanings, in the order the pieces appear.

HOW A NAME IS PARSED, STEP BY STEP

1. The tool removes a single trailing '.csv'.
2. The tool splits the segment suffix off. When the segment separator appears in the name, the tool decodes the text after its rightmost occurrence under the numbering scheme. It counts as a segment only when it decodes AND something is left in front of it. Otherwise the 'No number =' policy applies.

This step runs first and is permissive. It leaves a dotted value like '1.5' alone when the segment separator is '.'.

3. The tool tokenizes the rest on the separator.
4. The prefix, if any, matches the first token. The tool consumes it.
5. The tool walks the token order left to right against the remaining tokens. The optional fields behave like this:

dac, sample	consume the token unconditionally. A missing token is an error ("missing dac token").
ignore	consume a token if one is there.

pressure	consume the token ONLY if it reads as a number, after unit-stripping and decimal substitution. Otherwise the value stays at its default and the token goes to the NEXT field.
role	consume the token ONLY if it is a known channel keyword.
branch	consume the token ONLY if it is a known branch keyword.
rep	consume the token ONLY if it is all digits.

That conditional consumption lets one profile read vis_Y04_Arch29_bg.001 and vis_Y04_Arch29_26p0_bg_C_2.001 with one token order. An absent piece leaves its token to the next field.

- Any token left over is an error: "unrecognized trailing token".
- The resolved channel reads as dark, background or sample.

Defaults fill anything the name omits: value 0, channel dark, retake 1, plus the DAC and sample defaults you set.

TEACH BY EXAMPLE

Open 'Name format' from the left panel. The window holds the grammar boxes on top and the chip strip in the middle. The live whole-folder preview sits below them. A Guide card sits on the right.

- Pick a profile, or press 'Save as...' and name a new one. The built-in opens read-only. Saving under a new name starts you from an editable copy of it.
- Set the grammar boxes: separator first, then prefix, decimal, unit stripping, keywords, segment convention.
- Under 'Teach by example', pick a real filename from the folder. Pick one that shows EVERY piece your scheme can produce, so each piece gets a chip to label.

The tool decomposes the name in place. Each token gets a bold chip with a dropdown under it. Set the dropdown to what that piece means. The gray pieces between chips are the literal separators found in that name. Click one to jump to the Separator box. A matched prefix gets its own chip, whose dropdown reads 'prefix'. Change that dropdown to a field to label the piece like any other. The segment suffix sits at the end with a caption saying how the tool read it.

A tail can look like a segment number while the current settings skip it. The tool still shows it separately, with a caption naming the box to change ("segment tail: set Segment sep: '-"). Click it to jump to that box.

4. Watch the Preview. The tool parses every file in the folder with the grammar as it stands. Green rows parsed. Red rows skipped, with the reason in the note column. Blue rows fixed by hand. The counter under the list reads "matched N / M files". You can drag the columns. Hover a clipped cell to see it in full.

The preview reads at most the first 500 files in the folder. It says so when it stops there.

5. When the match count satisfies you, press 'Use this profile'. The tool saves the profile and Run uses it. The tool processes the folder when you press Run.

A grammar with a problem makes the button refuse, and the tool lists the problem. The problems are:

- an empty separator
- an unknown field
- a role keyword that maps to no channel
- dac or sample supplied by neither a token nor a default
- an unusable segment scheme
- a bad missing-segment value

GUESS FORMAT

'Guess format' reads the folder and proposes the whole grammar. It is a starting point. The preview is the real check.

What it does:

Segment convention first. It scores candidate separators (. - __seg -seg) against both numbering schemes. It looks for the same base recurring with SEVERAL different segment values (x.001 and x.002), which is the strong signal. Coverage alone is weak, because dotted values look the same. If every file it examined is numbered, it sets 'No number =' to 'reject'. If only some are, it sets 1.

Then the token separator, chosen for the most consistent token count across the folder.

Then a shared literal first token becomes the prefix. It skips that step when absorbing the token would leave fewer than two id columns. That happens in a single-cell folder where the shared token IS the DAC id. It then retries with no prefix.

Then it classifies columns: a value column (a token that parses

as a number, where bare integers alone are a weak signal), a channel-keyword column, a branch column, a retake column. It knows the usual words: bg / ref / back / background, s / sam / samp / sample / sig, dark / dk / drk, c / comp / up, d / dec / decomp / down.

The first two unclaimed columns become dac and sample.

It reports "matched N / M files" as a toast and in the log. It hands back a plain default in two cases. One is a folder with fewer than two id columns. The other is a profile the validator rejects.

WORKED EXAMPLES

Every piece present, dash-separated:

vis-D42-fo90-15.3GPa-s-c-2.003

Prefix	vis
Separator	-
Value decimal	.
Strip units	gpa
Sample keyword	s
Compression	c
Segment sep	.
Numbering	digits
Order	dac, sample, pressure, role, branch, rep

That reads as DAC D42, sample fo90, 15.3, sample channel, compression, retake 2, segment 3.

Letter segments (a/b/c):

ol_run7_4p2_bg_b

Segment sep	_
Numbering	letters
No number =	1

The rightmost '_' splits 'b' off, which decodes as segment 2.

Here the segment separator and the token separator are the same character. That works. It also means the tool reads any name ending in a one-letter or two-letter token as a segment. See the traps below.

Give segments their own separator when you can, such as '-b' or '_segb'. The two conventions then stay apart.

Zero-padded to two digits (.01):

Numbering 'digits' reads .01, .1 and .001 identically, because the tool parses the suffix with int().

Dash-numbered segments (-1, -2, -3) with underscore tokens:

quartz_C3_8p1_s-2

Separator	_
Segment sep	-
Numbering	digits

The two conventions stay apart, because '-' serves the segment alone here.

Single-cell folder, with DAC and sample outside the names:

12p5_bg.001

Prefix	(blank)
Separator	_
Default DAC Name	Y04
Default Sample Name	Arch29
Order	pressure, role

The dac and sample chips are absent from the order. The defaults supply both. The output CSVs are still named Y04_Arch29_12p5_absorbance.csv.

Names with a spare piece:

vis_20260731_Y04_Arch29_26p0_s.003

Label the date chip 'ignore'. The tool consumes and discards it.

EDGE CASES AND TRAPS

Decimal character equal to the segment separator.

Take decimal '.' with segment sep '.'. A name ending in a dotted value ('..._2.5') has its '5' read as a segment. The segment split runs first, on the character alone. The preview makes it visible at once: the value column reads 2 and the segment column reads 5. Give segments a different separator, or use 'p' for the decimal.

Token separator equal to the segment separator.

This is legal, and sometimes unavoidable. The cost is that any trailing token that decodes under the numbering scheme becomes the segment. Under 'digits' the tool eats a trailing retake number. Under 'letters' it eats a trailing one-letter or two-letter keyword. Check the preview's segment column against a name you know.

A value of 0.

It is allowed and meaningful. The tool also reads a MISSING value token as 0. For that group the log records "no pressure in filename, assumed 0 GPa", so the two cases stay apart.

Negative or non-finite values.

The tool takes a finite value at or above 0, and rejects the rest.

Case.

The tool matches prefix, channel keywords, branch keywords, unit suffixes and letter segments case-insensitively. It keeps DAC and sample ids exactly as written, because they become file names.

Raw/.csv twins.

The tool recognises a file and its .csv copy as one logical segment and counts it once. It logs a skipped name once as well.

Data extensions.

Under 'letters', a segment runs to two letters, so .txt / .dat / .asc survive as extensions. Under 'digits' the same holds, because those suffixes carry letters.

Two profiles, one folder.

One profile is active at a time. When a folder mixes two conventions, run the majority convention and fix the minority by hand, below. You can also split the folder.

FIXING STUBBORN FILES

A few files earn a hand fix of their own: the one hand-labelled retake, the stray export with a date in the middle. Double-click any row in the preview to type its fields by hand. Selecting it and pressing 'Fix selected...' does the same.

Channel role dark / background / sample
DAC
Sample
<Series variable>
Replicate
Segment

Leave a field blank to keep what the parser produced. A fix beats any pattern. A fix that supplies a channel role can even resurrect a file the parser rejected outright. A fix on a rejected file needs that role, and the tool ignores a fix that leaves it blank.

'Exclude selected' leaves the file out of the run entirely. The

preview note then reads "excluded by user". 'Remove fix' inside the fix dialog undoes one file. 'Clear all fixes' forgets every fix and exclusion for the folder.

The tool stores fixes per folder and keeps them across restarts. It applies them after parsing, on every Run, Rescan and preview.

WHERE THE PARSED VALUES GO

dac, sample and the value become the output file name stem: {DAC}_{SAMPLE}_{VALUE}[C|D]_absorbance.csv, with 'p' as the decimal.

The trace's identity label is "{DAC} {SAMPLE} {VALUE} GPa", plus a branch tag and a retake note. The interface relabels it to the active Series variable's unit for display. The identity label itself holds. It is the key that presets, sessions and exports resolve through.

The channel decides which of Sample / Background / Dark the file contributes to. The segment decides concatenation order. The retake decides which measurement wins, which is the highest index.

WORKFLOWS

Recipes by goal. Start from the thing you need and work down. Each recipe assumes a folder is loaded. To rehearse a recipe first, point the Input folder at the bundled demo dataset and follow it through there.

A PUBLICATION OVERLAY FIGURE

1. Run the folder. Confirm the trace count in the Progress log matches what you measured.
2. DATA > TRACES: tick the traces for the figure. Leave 'Lock colors to all datasets' on in STYLE > COLORS & COLORMAP. The colors then hold as you untick.
3. EXPORT > FIGURE: pick the journal preset for the column you are filling. That sets width, typeface, sizes, line weight, spines and DPI in one action.
4. EXPORT > EXPORT: tick 'Preview at export size (WYSIWYG)'. Judge the figure at that size. 7 pt type at 89 mm is smaller than it looks in a full window.
5. Decide the key. Up to about ten traces, use STYLE > LEGEND with Location 'outside right'. Above that, use STYLE > COLORBAR, or 'Direct labels at curves' in STYLE > LEGEND. Direct labels usually read best.
6. AXES > LIMITS & SCALE: set the X range you want to show. A zoom on the plot fills these boxes for you.
7. STYLE > TITLE & AXIS LABELS: the axis labels default sensibly. Override them when the journal has a house form. Mathtext works: λ (nm), Fe^{2+} .
8. Save the plot as PDF with 'Editable text' on. Tick 'Also save' PNG for slides. Add 'Grayscale copy' when the journal prints some figures in mono.
9. The export may auto-fit the legend. The tool writes the columns and font size it used back into the panel. Copy them to make that layout permanent.
10. Left panel > Progress > 'Export settings' prints the whole configuration into the log for the methods section.

DEFRINGED CSVs FOR IGOR OR ANOTHER TOOL

1. Tick df on the strip beside the plot. That is the whole first pass. The tool finds the strongest ripple in each channel and takes it out.
2. Read the fringe report in the log. It names the traces with a detected fringe, the fitted $n \cdot t$ in micron, and the p-value. An

absent trace is one the detector missed, and it passes through unchanged.

3. Sanity-check one trace. In PLOT > 2D PLOT OPTIONS, tick 'Defringe compare (selected trace)', then click a curve. The gray dashed line behind it is the pre-defringe absorbance. If the notch took out structure you care about, narrow the half-width in FRINGE > FFT REMOVAL > Notch list. If fringes survive, widen it, or loosen 'Fisher p' under FRINGE > PANELS > Detection. Press 'Write to defringe' to send your settings to the export.
4. EXPORT > EXPORT: pick 'Export CSV...' > 'Defringed CSV (FFT-notch absorbance)'. Tick Crop and give a wavelength range first to export part of the spectrum.
5. Pick a destination folder. You get one {stem}_absorbance_notch.csv per trace. The columns are Wavelength, Dark, Background, Sample, Absorbance. A channel the tool notched adds Background_notch, Sample_notch and Absorbance_notch. An un-notched channel leaves its notch column blank, so a reader can tell what the tool touched.

Leave df ticked and every Run writes the notch files beside the absorbance CSVs.

SMOOTHED CSVs

Take the same path with 'Export CSV...' > 'Smoothed CSV (raw + smoothed columns)'. You get wavelength, wavenumber, eV, the raw columns and the smoothed ones side by side. Tune the filter first in DATA > SMOOTHING > 'Smoothing settings...', with live preview on. The split at 600 nm gives the two grating regions their own windows.

THICKNESS VERSUS PRESSURE

1. Tick df beside the plot. The detection supplies n^*t .
2. Check the fringe report. A thickness point needs a confident detection. The 'mark misses' triangles show the rest.
3. PLOT > PLOT MODE: switch to Thickness (fringe n^*t). Sample points are filled circles and background points are open squares. The S / B boxes pick the channels.
4. For solved physical thickness, take the traces through the fringe workbench: describe the stack, assign roles, Solve, Record point. 'Results vs pressure...' in FRINGE > SESSION plots the solved values. Turn on 'Error bars (multiscale variance)' before you quote a number.
5. 'Thickness table...' in PLOT > PLOT MODE lists n^*t and its

p-value per trace. The plot exports through the normal Save plot path.

COMPARING COMPRESSION AND DECOMPRESSION

1. Set the branches. An explicit `_C / _D` tag in the filename wins. The tool recognises ten historical experiments automatically. Otherwise tick the D box per trace in DATA > TRACES, or load a list.
2. 'Decompression list...' in DATA > TRACES reads a CSV or text file of decompression values. It takes a pressure_GPa column or plain numbers, allows 'p' as the decimal, and ignores headers. Each value takes the nearest loaded trace within 0.05. The tool remembers the list per cell and sample.
3. Read one branch at a time with 'Only C' and 'Only D' before you overlay them.
4. Overlay both. The Decompression traces controls in PLOT > 2D PLOT OPTIONS draw the D branch dashed under 'Style D traces apart'. Give it its own width, opacity and markers when dashing alone separates the branches too little.
5. STYLE > LEGEND: rename 'C' and 'D' in the two small boxes. Other words may fit your experiment better, such as 'heat' / 'cool' or 'inc' / 'dec'. That rename is display only. The file names and the D list keep C and D.
6. 'Export D list (CSV) by selection' writes the branch assignment back out, so the next person has it.
7. 'Save C/D-tagged CSVs...' writes the whole set with the branch letter in every file name.

A CUSTOM QUANTITY, PLOTTED AND EXPORTED

1. DATA > FORMULAS: press New...
2. Name it and give it a unit. The name labels the Y axis and names the CSV column.
3. Write the expression over the loaded columns. The tool derives the typeset preview under the box from what you typed, so the two always agree. The problems list and a live min / max / NaN preview update as you type. Save stays disabled until the formula is clean.
4. Tick df first when the formula uses Sf, Bf or Af. Turn 'Show smoothed' on when it uses As. The tool reports a missing column and leaves a gap.
5. Save, then click the dot beside the row. The formula becomes the Y axis ('formula: <name>'). The row tints and goes bold.

6. Style and export the figure as you would for absorbance.
7. 'Save formula CSVs...' writes one two-column file per trace: wavelength plus the formula, with the expression in the header comments. One provenance sidecar covers the batch. The absorbance CSVs stay untouched.

Worked example, what the smoother removed:

```
expression A - As
unit      (blank)
needs     Show smoothed on
Plot it and you get the residual. The residual is what decides
whether a smoothing setting is defensible.
```

LIVE ACQUISITION AT THE BEAMLINE

1. Run once as soon as there are a few measurements. The tool then has a baseline file list.
2. PLOT > PLOT MODE: turn on Auto rescan. Set the interval to roughly your measurement cadence.
3. Leave it. The status line under the plot reads 'auto-rescan: N s' while the timer is armed. A tick that finds new files re-reduces the folder and says so in a brief toast. The timer fires while the tool sits idle.
4. Rescan and F5 still work by hand, mid-run included.
5. Use Inspect one trace on each new point as it appears. That is the moment to notice a saturated channel or a drifting lamp.

COMPARING TWO DATASETS SIDE BY SIDE

1. Press Ctrl+T for a new session tab. Each tab has its own data, folders, settings and undo history.
2. Load or Run the second dataset there. The tab names itself after the folder.
3. Ctrl+Tab cycles. Style each tab on its own. You can also style one tab and save it as a preset (EXPORT > PRESETS & PROJECTS). Load that preset in the other tab for identical styling.
4. Double-click a tab to rename it something you will recognise in an hour.

COMING BACK TO A SESSION LATER

Three levels. Pick by how much you need back.

The curves alone: 'Load previous run...' on the left panel. Pick the

output subfolder a Run wrote. The tool re-imports its *_absorbance.csv files for plotting, straight off disk. Smoothing and defringe still apply live on top. A blank session tab also lists recent runs for one click.

The styling: EXPORT > PRESETS & PROJECTS, pick the preset and Load. A preset carries the plot controls alone, so one preset styles many datasets.

Everything: 'Open project...'. A project is every setting plus the input and output folders, in one .json. It reopens an analysis exactly where you left it.

The tool remembers the rest between launches: folders, theme, window size, the default colormap, your notes, and your presets.

KEYBOARD SHORTCUTS

Files and figures

Ctrl+S	Save plot
Ctrl+Shift+C	Copy figure to clipboard
F5	Rescan the input folder (top-up run)
Enter (input)	Rescan too, and remember that folder

Editing

Ctrl+Z	Undo
Ctrl+Y	Redo (also Ctrl+Shift+Z)

Session tabs

Ctrl+T	New tab (blank session)
Ctrl+W	Close the current tab
Ctrl+Tab	Next tab (Ctrl+Shift+Tab = previous)

The plot, with the mouse

Click curve	Select it (highlight; defringe-compare target)
Dbl-click	Solo that curve (again = restore)
Click legend	Hide / show that trace
R-click curve	Quick-actions menu: inspect this trace, solo, hide, toggle decompression (D), defringe compare here, show in data table, deselect
Drag a box	Zoom into it (2D)
Mouse wheel	Zoom (2D about the cursor; 3D camera)
Dbl-click pill	Reset that pane divider to its default width

The plot, with the keyboard

Arrow keys	Pan the 2D view / orbit the 3D camera (3 deg a step)
+ / -	Zoom (2D about center; 3D camera)
0	Fit the 2D view / reset the 3D camera
1 / 2 / 3 / 4	Stacked & 3D: off / 2D stacked / 3D ridge / 3D shape
[/]	Previous / next colormap
Ctrl+R	Reset 3D view

Data

Ctrl+D	Toggle the raw data table
Ctrl+C (table)	Copy the selected data-table rows

Elsewhere

Drop a folder	Make it the input folder and rescan (Windows)
F1	This shortcuts list
Esc	Close the dialog on top

The tool ignores single keys while you type in a box. Single keys are the numbers, the brackets, the arrows, +/- and 0. Escape also cancels a pane-divider drag.